

DistriCine: Seven Art Make Code

Diego Alejandro Yañes Zabala (20251020103)
Brayan Sebastian Diaz Ramirez (20251020150)
Oscar Javier Camargo Trujillo (20251020143)

*Francisco José de Caldas District University
Faculty of Engineering
Systems Engineering Program
Bogota DC, Colombia*

Teacher Eng. Carlos Andres Sierra, M.Sc

Abstract—DistriCine is a specialized application designed specifically to manage university film clubs at the Francisco José de Caldas District University. The system addresses the problem of low student participation caused by manual and non-digital processes, offering a comprehensive solution based on Object-Oriented Programming (OOP) principles. The application implements essential functionalities such as viewing a film catalog with complete metadata, authentication with institutional email, detailed access to film information, and reservations by location and time. Developed in Java with a three-tier modular architecture, DistriCine demonstrates how a robust OOP-based design allows for the creation of scalable and maintainable systems for the university context, using JSON file persistence and a JavaFX graphical interface that meets all implemented functional requirements.

1. Introduction

At the Francisco José de Caldas District University, university film clubs face a significant challenge: low student participation. Cultural activities are coordinated primarily through traditional methods—physical posters, classroom announcements, and word-of-mouth communication—which limits their reach and effectiveness [1]. An analysis conducted in the second half of 2025 revealed that less than 15% of students were aware of scheduled screenings, and average attendance was less than 10 people per event.

DistriCine was created as a direct response to this problem. Unlike commercial platforms such as CineMark or Eventbrite, which are complex, expensive, and designed for commercial environments, DistriCine has been developed specifically for the university context, integrating essential features such as:

- Authentication with university institutional email
- Management of multiple locations (campuses)
- Clear differentiation of roles (student and administrator)
- Data persistence adapted to academic environments with limited resources

The development of DistriCine has followed an approach focused on architectural robustness rather than visual aesthetics, prioritizing the rigorous application of OOP principles—encapsulation, inheritance, polymorphism, and composition—as the foundation for a maintainable and extensible system [2]. This approach contrasts with many academic prototypes that prioritize visually appealing interfaces but with fragile architectures [3].

This article presents the current implementation of DistriCine, detailing its object-oriented architecture, the design decisions that enable its efficient operation, and the results obtained after its functional validation. The system represents a concrete, reproducible solution specifically adapted to the cultural management needs of university environments.

2. Methods and Materials

DistriCine was developed through an iterative process that began with the precise identification of requirements and culminated in a validated, functional prototype. The system was implemented in Java 17 with a JavaFX graphical interface, maintaining a clearly defined three-layer architecture (Presentation, Business Logic, and Data Access).

2.1. Architecture and Key Components

The heart of DistriCine is its object-oriented domain model. Figure 1 shows the class diagram that represents the fundamental structure of the system:

Main Classes:

- **User (abstract)**: Defines common attributes (name, institutional email, password) and polymorphic behavior through the abstract method `showMenu()`.
- **CustomerUser**: Extends User for students, implementing functionalities such as catalog viewing and making reservations.
- **AdminUser**: Extends User for administrators, allowing management of movies and schedules.

modifications to existing classes, validating the effective application of SOLID principles.

Figure 3 shows a sequence diagram illustrating the complete flow of a booking, demonstrating how DistriCine's components collaborate consistently to meet this fundamental use case.

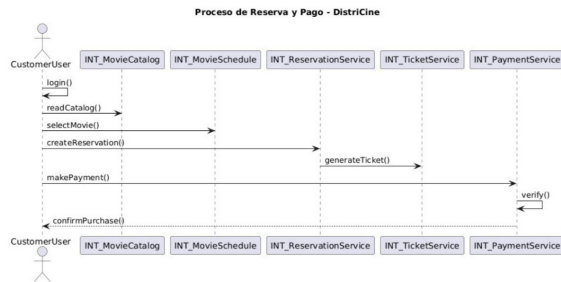


Figure 3: Sequence diagram of the booking flow at DistriCine

4. Conclusions

DistriCine represents a concrete and effective solution for managing university film clubs at the Francisco José de Caldas District University. The system demonstrates how a robust application based on Object-Oriented Programming principles can address specific cultural challenges in the academic sphere.

The results obtained validate that DistriCine successfully fulfills its main objectives:

- 1) To provide a specialized platform for university film clubs
- 2) Facilitate access to information on films and screening times
- 3) Simplify the booking process for students
- 4) Maintain persistent information between usage sessions

The modular, object-oriented architecture proved fundamental to the project's success, enabling a clear, maintainable, and extensible implementation. The persistence strategy using JSON files was well-suited to the university context, eliminating installation and configuration barriers.

As future work, the following improvements are planned for DistriCine:

- Integration with institutional authentication systems for Single Sign-On
- Implementation of filtering by gender and other criteria
- Development of a reporting module for administrators
- Migration to a relational database for better scalability
- Implementation of automatic user notifications

DistriCine is not simply a technical application—it is a tool designed to revitalize film culture at the university,

making it easier for more students to discover and participate in these educational experiences. Its success demonstrates that well-designed solutions, focused on specific needs and built with technical rigor, can have a significant impact on academic cultural life.

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